FINITE ELEMENT ANALYSIS

III B. Tech. II Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME35	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

The aim of this course to enable students to understand fundamental theory of the FEA method, generate the governing FE equations for systems governed by partial differential equations, understand the use of the basic finite elements for structural applications using truss, beam and plane elements under static and dynamic loading; understand the application and use of the FE method for heat transfer problems.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Demonstrate basic concepts of FEM, Solve the basic spring problems
2. Formulate and solve finite element equations for 1-D elements and Trusses
3. Derive stiffness matrices, analyze beams and 2D CST Elements subjected to different load conditions
4. Formulate and solve axi-symmetric problems using constant strain triangular elements and heat transfer problems,
5. Analyze axial bar, beam and truss subjected to dynamic loads

UNIT-I — INTRODUCTION TO FEM

Basic concept, historical background, applications of FEM, Basic equations of elasticity-strain-displacement relations and stress-strain relations, Convergence requirements, Principle of Minimum Potential Energy-Spring Problems.

UNIT-II — ONE DIMENSIONAL PROBLEMS & ANALYSIS OF TRUSSES

ONE DIMENSIONAL PROBLEMS: Formulation of Stiffness Matrix for a Bar Element by the Principle of Minimum Potential Energy, Properties of Stiffness Matrix, Characteristics of Shape Functions, Quadratic shape functions. Problems on uniform stepped and tapered bars subjected to point loads, body force and temperature loads.

ANALYSIS OF TRUSSES: Local and Global co-ordinates, Transformation matrix, Stiffness Matrix for Plane Truss, Stress-Strain Calculations and Calculation of reaction forces.

UNIT-III — ANALYSIS OF BEAMS & 2-D PROBLEMS

ANALYSIS OF BEAMS: Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, slope and deflection. Simple problems-Point loads & UDL.

2-D PROBLEMS: CST Element-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems. Shape functions of Two dimensional four noded isoparametric element.

UNIT-IV — AXI-SYMMETRIC SOLIDS & HEAT TRANSFER

AXI-SYMMETRIC SOLIDS: Axi-symmetric solids subjected to Axi-symmetric loading with triangular elements-simple problems.

STEADY STATE HEAT TRANSFER ANALYSIS: One Dimensional Finite Element analysis of fin and composite slabs.	
UNIT-V	**DYNAMIC ANALYSIS**
DYNAMIC ANALYSIS: Dynamic Equations of motion, Lumped and consistent mass matrices, Evaluation of Eigen values and Eigen vectors for a stepped bar and beam.	
Text Books:	
1. Introduction to Finite Elements in Engineering by Chandrupatla, Ashok and Belegundu, Prentice, Hall, Pearson. 2. The Finite Element Methods in Engineering by SS Rao, Pergamon.	
Reference Books:	
1. Finite Element Methods: Basic Concepts and applications by Alavala, PHI 2. Finite Element Analysis Revised and Enlarged Edition by S.Md.Jalaluddin, Anuradha Publications 3. Finite Element Analysis by P.Seshu, PHI 4. Finite Element Analysis by Bathe, PHI 5. Finite Element Method by Krishna Murthy, TMH 6. An Introduction to the Finite Element Method (McGraw-Hill Mechanical Engineering) 3rd Edition. by J Reddy, McGraw-Hill	